

SafeSip: Real-Time Detection of Invisible Water Contaminants

Saatvik Sunilraj

Yash Sreepathi

Abstract

Unsafe drinking water is a critical global health crisis, with over 2.1 billion people lacking access to safely managed resources and an estimated 1.4 million annual deaths attributed to water contamination. The most dangerous ionic contaminants, including Lead, Arsenic, and Nitrates, are invisible, odorless, and tasteless. Current detection methods are inadequate: traditional laboratory-grade detection is cost-prohibitive (\$200-\$600/sample) and consumer test strips often yield detection limits up to 1,300x too high for safety standards. This study introduces SafeSip, a low-cost, mobile-integrated platform that utilizes multi-frequency Electrochemical Impedance Spectroscopy (EIS) and machine learning to identify specific ionic “fingerprints.” By sweeping frequencies from 1 kHz to 100 kHz through a custom 62-finger interdigitated PCB sensor, the system captures 40 distinct real (Z') and imaginary (Z'') impedance features. To overcome the “Water Matrix Problem,” where background minerals interfere with trace signals, this research developed and validated the Delta Method ($\Delta Z = Z_{\text{sample}} - Z_{\text{pure}}$), which isolates contaminant-specific signatures. Experimental results across a novel library of 500 samples demonstrate that a Random Forest classifier achieves a mean accuracy of 96.2% across 5-folds and a Macro F1 score of 0.9616. Furthermore, high Pearson correlation coefficients ($r = 0.9904$ real, $r = 0.9821$ imaginary) confirm that these “fingerprints” exhibit matrix invariance, maintaining integrity across water sources such as distilled and high-mineral matrices. SafeSip provides a scalable solution for under \$40 in hardware costs by being integrated with a cross-platform mobile application, including a real-time contamination heat map. SafeSip moves beyond simple detection to democratize access to water quality detection by empowering communities to monitor invisible threats, inform targeted interventions, and strengthen accountability in regions facing water insecurity.

Introduction

Access to safe drinking water remains one of the most urgent global health challenges. Over 2.1 billion people worldwide lack access to safely managed drinking water services, and approximately 1.4 million deaths each year are attributed to waterborne contamination (WHO/UNICEF Joint Monitoring Programme, 2025; Wolf et al., 2023). Hazardous contaminants such as lead, arsenic, and nitrates are particularly dangerous because they are colorless, odorless, and tasteless, rendering them undetectable without specialized instrumentation (WHO, 2024; WHO, current; Temkin et al., 2019). Long-term exposure can lead to neurological damage, developmental disorders, and increased cancer risk (Lanphear et al., 2005).

Existing detection methods remain inadequate. Laboratory-based testing is highly accurate but costs between \$200 and \$600 per sample and often requires up to two weeks for results, making it inaccessible in many resource-constrained settings (Kruse, 2018; Bain et al., 2021). Consumer test strips, while inexpensive, are limited in scope and reliability. Studies have shown that many commercially available kits fail to meet safety thresholds, with detection limits up to 1,300x higher than acceptable drinking water standards (Kriss et al., 2021). This gap between accuracy, affordability, and accessibility demonstrates the need for a fundamentally new approach to water quality monitoring.

A potential solution lies in the electrical properties of water. Dissolved ionic contaminants conduct electricity differently than pure water (Zhang et al., 2020; Hasted et al., 1948). When an alternating current (AC) is applied, ions respond uniquely to the oscillating electric field, with responses varying according to ion size, mobility, and charge density (Persson, 2017). Electrochemical Impedance Spectroscopy (EIS) exploits this phenomenon by sweeping across multiple frequencies, generating a spectral “fingerprint” for each contaminant class (Armaki et al., 2025; Opekar et al., 2013). Unlike conventional conductivity measurements that only

indicate dissolved material, EIS provides frequency-dependent information capable of differentiating ionic species.

The real component $Z'(f)$ reflects the resistive properties of the solution, while the imaginary component $Z''(f)$ captures capacitive behavior associated with electrical double layer formation at the electrode surface (Khademi & Barz, 2020). Together, these components create a contaminant-specific electrochemical signature.

However, applying EIS in real-world conditions introduces a major challenge known as the water matrix problem. Natural water sources contain complex backgrounds of dissolved minerals and ions that can distort target contaminant signals, reducing measurement reliability (Bain et al., 2021; UNEP GEMS/Water, 2025). Any deployable system must therefore achieve matrix invariance so that contaminant detection remains stable across varying water compositions.

To address these challenges, this study introduces SafeSip, a low-cost, mobile-integrated sensing platform designed for ionic contaminant detection through electrochemical impedance analysis. Three primary engineering objectives guided the system design: accuracy, affordability, and accessibility. SafeSip utilizes a custom interdigitated PCB sensor, a programmable impedance analyzer, and a machine learning classification framework to interpret complex electrochemical patterns. By extracting real and imaginary impedance features across multiple frequencies, the system generates high-dimensional electrochemical fingerprints that can be classified using supervised learning models.

Beyond detection, SafeSip incorporates a mobile application and contamination heat map that aggregate geotagged user data into a real-time environmental monitoring network. This enables communities and public health organizations to identify contamination hotspots and monitor trends over time, transforming isolated measurements into actionable environmental intelligence.

METHODS AND MATERIALS

SafeSip was engineered as an integrated sensing and analysis system designed to achieve three primary objectives: accurate contaminant classification, affordability for resource-constrained deployment, and accessibility through smartphone integration.



Figure 3.1

Figure 3.2

Figure 3.3

The sensing platform was built around a custom interdigitated electrode array fabricated using printed circuit board (PCB) architecture. As shown in Figure 3.1, the sensor consists of 62 interdigitated copper fingers arranged across a 30×30 mm active area with 0.2 mm trace width and spacing. This geometry maximized the electrode-electrolyte interface, increasing sensitivity to ionic interactions and amplifying measurable impedance responses.

Electrochemical impedance measurements were generated using an AD5933 impedance analyzer connected to an ESP32 microcontroller through I2C communication. The AD5933 produced an alternating current signal across a frequency sweep from 1 kHz to 100 kHz. For each sample, the ESP32 initiated a full frequency sweep and recorded impedance responses, generating 40 total features per sample consisting of 20 real Z' values and 20 imaginary Z'' values. The complete acquisition setup is shown in Figure 3.2.

All components were integrated into a compact handheld enclosure fabricated using PLA-based 3D printing, producing a portable and field-deployable device (Figure 3.3). Total hardware cost was maintained under \$40, including the ESP32 microcontroller (\$10), AD5933 impedance

analyzer (\$23), interdigitated PCB sensor (\$1.04), and supporting circuitry such as resistors, breadboards, and wiring (\$5.13). This low-cost design supports scalable deployment.

A total of 500 water samples were prepared, consisting of six ionic contaminants and three simulated real-world mixtures (Figure 4). To isolate contaminant-specific electrochemical fingerprints independent of signal magnitude, all solutions were standardized to 200 ppm (mg/L). Samples were prepared using a 4-decimal analytical balance (± 0.1 mg) and Class A volumetric glassware. Each compound was dissolved in 100 mL of baseline water to create a 2000 ppm stock solution, then diluted 1:10 to the target concentration. Liquid concentrates were pipetted to constant volumes for consistency. Binary mixtures were prepared in a 1:1 ratio to simulate environmental contamination scenarios such as agricultural runoff and industrial waste. To reduce bias from temporal drift and environmental fluctuations, a randomization algorithm determined sample testing order.

During measurement, water samples were applied directly to the interdigitated sensor, forming a conductive path between electrodes. The system applied alternating current while sweeping frequencies, capturing resistive and capacitive responses associated with ion transport and electrical double layer formation. Due to instability at the lower operating boundary of the AD5933, the two lowest frequencies (1.0 kHz and 1.274 kHz) were excluded from analysis. This produced 18 stable frequencies spanning 1.624 kHz to 100 kHz and 36 total features per sample consisting of paired real $Z'(f)$ and imaginary $Z''(f)$ impedance values.

To address the water matrix problem, where background ions distort contaminant signals, a preprocessing technique known as the Delta Method was applied. For

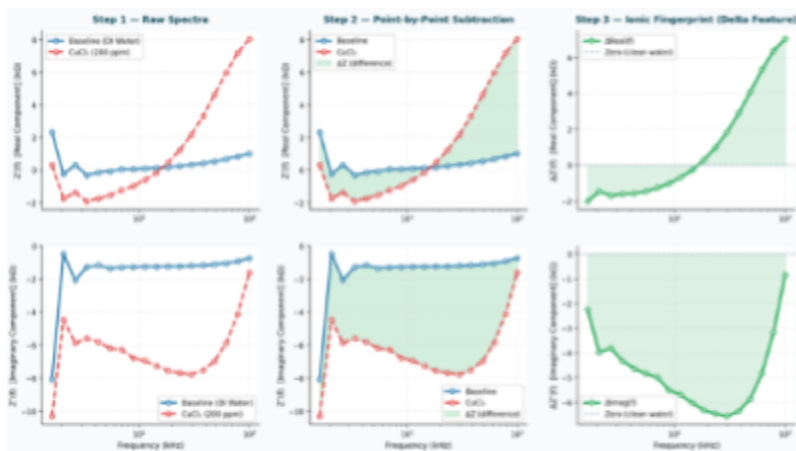


Figure 5

each frequency point, the mean impedance spectrum of pure baseline water was subtracted from the measured sample, producing transformed features $\Delta Z'(f)$ and $\Delta Z''(f)$ (Figure 5). This isolated contaminant-specific electrochemical variations and improved consistency across varying water compositions, enabling matrix-invariant detection.

Processed feature vectors were used to train a supervised Random Forest classifier consisting of 500 decision trees. Model performance was evaluated using stratified 5-fold cross-validation, where each fold reserved 100 real samples for testing and trained on the remaining 400. To simulate sensor variability and improve generalization, Gaussian noise augmentation was applied only within training folds, with noise scaled to per-feature standard deviation. This expanded each training fold to 500 samples while preventing augmented data leakage into the test set.

SafeSip was integrated with a cross-platform Flutter mobile application that enabled real-time interaction between hardware and user interface. The application communicated with the ESP32 through Bluetooth Low Energy (BLE) or USB connection. After initiating a test, impedance data were transmitted to the application, where baseline subtraction and feature processing were performed. The processed data were then analyzed using the on-device Random Forest classifier, producing contaminant classification and confidence scores in under four seconds.

Results were displayed with a safety indicator and could optionally be uploaded to a Firestore-backed database, where each measurement was geotagged and incorporated into a live contamination heat map (Figure 6).



Figure 6

Results

SafeSip demonstrated strong performance in classifying ionic contaminants using electrochemical impedance-derived feature vectors. Across stratified 5-fold cross-validation, the Random Forest classifier achieved a mean accuracy of 96.2%, indicating high reliability in distinguishing contaminant classes (Figure 7). The Macro F1 score of 0.9616 further confirms balanced performance across all classes, demonstrating that the model minimized both false positives and false negatives while maintaining predictive capability across individual contaminants and mixed samples.

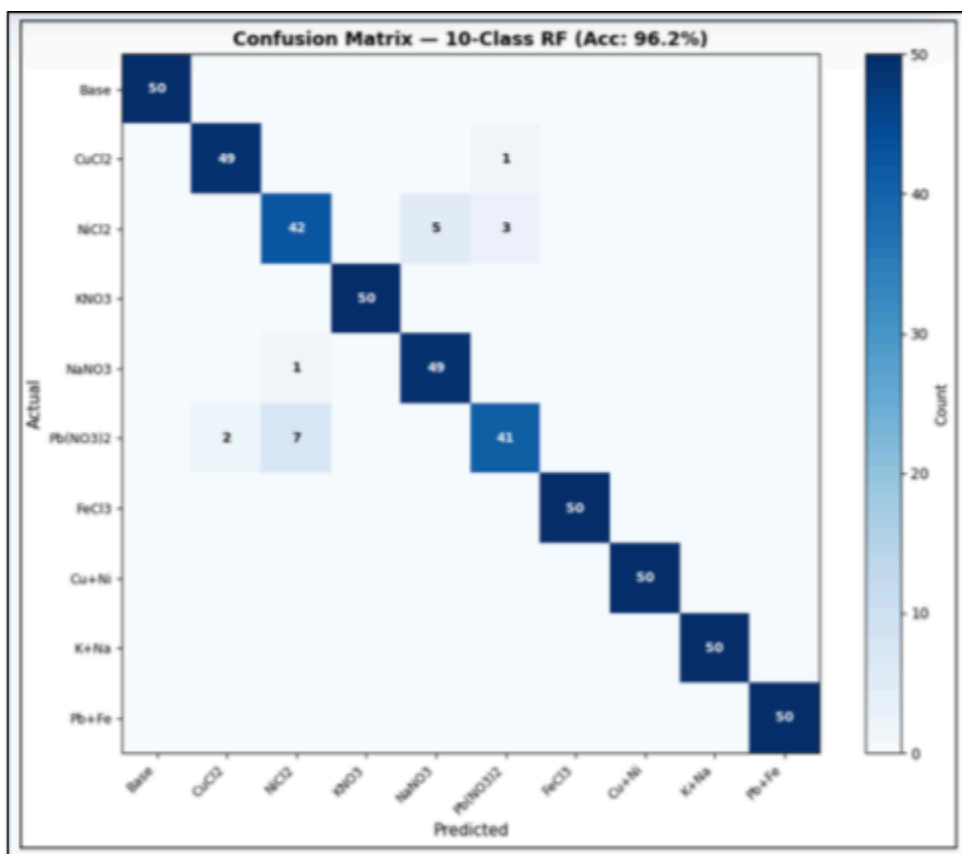


Figure 7

Analysis of classification outputs showed that electrochemical impedance spectroscopy (EIS) features provided sufficient discriminatory power to differentiate contaminants with similar bulk

electrical properties. Although all samples were standardized to the same concentration (200 ppm), the model correctly identified contaminants using frequency-dependent impedance variations rather than signal magnitude alone. This confirms that the extracted features captured ion-specific electrochemical behavior, validating multi-frequency impedance as a fingerprinting approach.

To evaluate robustness under varying environmental conditions, model performance was analyzed across different water matrices, including distilled and high-mineral compositions. Strong Pearson correlation coefficients between impedance spectra ($r = 0.9904$ for real components and $r = 0.9821$ for imaginary components) showed that fingerprint structure remained preserved across matrix conditions. These findings demonstrate that the Delta Method effectively mitigated background interference, allowing contaminant-specific signals to remain consistent in the presence of competing ions (Figure 8). This matrix invariance is critical for real-world deployment, where water composition is highly variable.

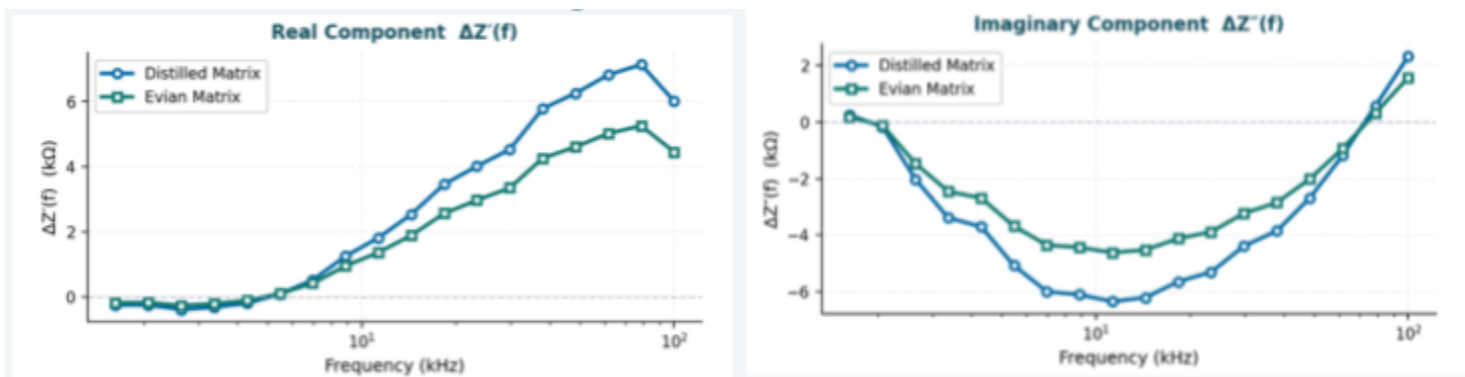


Figure 8

Binary mixture samples further tested the system's ability to handle complex contamination scenarios. The classifier maintained high accuracy when distinguishing mixed samples from single-contaminant solutions, suggesting that the model learned overlapping electrochemical patterns rather than isolated signals alone. This demonstrates the system's capacity to operate in realistic environments where multiple contaminants may coexist.

System performance was also evaluated in terms of speed and deployability. SafeSip delivered contaminant identification and confidence scoring in under four seconds per sample through the mobile application interface. This represents a major improvement over traditional laboratory testing, which often requires days to weeks for results. The lightweight feature representation and efficient Random Forest architecture enabled on-device inference without cloud-based computation, supporting real-time field deployment.

The mobile integration component further extended system functionality by enabling spatial aggregation of test results. Measurements were successfully transmitted, geotagged, and visualized within a live contamination heat map. This transformed individual measurements into a broader environmental monitoring network capable of identifying contamination hotspots and temporal trends. The ability to collect and visualize distributed water quality data in real time addresses a major limitation in existing monitoring systems, which are often geographically sparse and infrequently updated.

Overall, the results demonstrate that SafeSip achieves accurate, rapid, and scalable detection of ionic water contaminants. The integration of electrochemical sensing, machine learning classification, and mobile-based data aggregation provides a technically robust and practically deployable solution for real-world water quality monitoring.

Discussion and Conclusions

The results of this study demonstrate that SafeSip effectively addresses major limitations in existing water quality monitoring systems through the integration of electrochemical sensing, machine learning, and mobile-based visualization. The high classification accuracy and balanced Macro F1 score indicate reliable performance across both individual contaminants and mixed samples. Because all samples were standardized to the same concentration, the model's performance confirms that classification was driven by frequency-dependent electrochemical behavior rather than simple signal magnitude differences. This validates multi-frequency electrochemical impedance spectroscopy (EIS) as a robust fingerprinting mechanism for ionic detection.

One of the most significant contributions of this work is the effectiveness of the Delta Method in achieving matrix invariance. Strong correlations between impedance fingerprints across distilled and high-mineral water matrices demonstrate that baseline subtraction successfully isolates contaminant-specific signals from background interference. By maintaining consistent performance across varying water compositions, SafeSip overcomes a major barrier to real-world deployment of electrochemical sensing systems.

The successful classification of binary mixtures further supports the system's practical applicability. Since real-world contamination rarely occurs in isolation, the ability to identify mixed samples suggests that the model learned overlapping electrochemical signatures rather than isolated signals alone. This demonstrates strong generalization capability and reinforces the value of combining physics-based sensing with data-driven classification. Additionally, SafeSip achieved contaminant classification in under four seconds, representing a substantial improvement over traditional laboratory testing that often requires days to weeks. The lightweight feature representation and efficient Random Forest architecture enabled real-time,

on-device inference without cloud computation, supporting deployment in low-resource environments.

Beyond detection, the integration of a mobile application and geospatial heat map significantly expands the system's impact. By converting individual measurements into GPS-tagged data points, SafeSip enables real-time visualization of contamination patterns and hotspots. This addresses a major limitation in current monitoring systems, which are often geographically sparse and infrequently updated, while providing actionable insights for communities, non-governmental organizations, and public health agencies.

Despite these strengths, several limitations should be acknowledged. The dataset was limited to six contaminants and a defined set of binary mixtures at a fixed concentration. Expanding the dataset to include additional contaminants, concentration ranges, and more complex multi-ion interactions would improve model robustness and generalizability. Although Gaussian noise augmentation was used to simulate sensor variability, additional validation under uncontrolled field conditions remains necessary. Environmental factors such as temperature fluctuations, sensor fouling, and long-term drift may affect performance and should be explored in future work.

In conclusion, SafeSip demonstrates that low-cost hardware, electrochemical sensing, and machine learning can be combined into a scalable, real-time water quality monitoring system. The platform achieves high accuracy, rapid response times, and consistent performance across varying water matrices while remaining affordable and accessible for deployment in resource-constrained environments. By enabling community-level data aggregation and visualization, SafeSip supports decentralized, data-driven water monitoring that can improve access to water quality information and help reduce risks associated with unsafe drinking water.

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